

MedWorld-JEPA: Learning a Medical World Model from Longitudinal Observations

Supplementary Material

A. Evaluation Details

A.1. Longitudinal inputs and readouts

The current cohort construction pairs adjacent frontal MIMIC-CXR studies from the same admission, with follow-up intervals of 6–72 hours and requested horizon bins of 6–24 and > 24–72 hours. Linked MIMIC-IV history is restricted to the source patient and admission, with both event and recording times no later than the source acquisition. The current radiology report is assumed available retrospectively because reliable signing times are unavailable. Future observations supply training targets and scoring references only.

In the completed legacy prototype, a text decoder reads the predicted future state to generate the follow-up report, and a finding head supplies continuous scores. Forecast training uses 16,000 pairs, validation uses 230, and testing uses 297 pairs from 94 patients. The full-token, eight-slot, and shuffled-state conditions use Qwen3.5-9B. All three complete 2,400 forecast updates at effective batch size 32 (seed 42) from a common newly trained 9B Stage 1 initialization, with metrics and GREEN scored on all 297 test pairs. The slot adaptation extracts eight learned queries from the final language layer; the full-token condition retains valid image/report/history tokens. Both retain the forecast module. The shuffled condition permutes source states across patients. It generates one unique report across all 297 pairs, compared with 224 for slots and 287 for full-token. These adaptations are separate from the proposed multi-depth 4 + 4 design; archived 0.8B scores do not populate their rows.

Revised native forecast protocol (pending). The new Qwen conditions preserve the native vision encoder and language decoder with the same 512-pixel, aspect-preserving current image, truncated current report, source-available EHR, and requested horizon. The state branch constructs four JEPA–language fusion slots and four native-vision slots, each of width 1024; a horizon-conditioned predictor outputs eight future slots. Their projection is inserted at the end of the native user message, before the assistant prompt. The native supervised baseline omits these state tokens; the augmented and shuffled conditions retain them. Only the shuffled state branch reads another patient’s source observation from the same split, while the decoder’s true current image, report, EHR, and horizon remain unchanged. The true follow-up enters the frozen Stage 1 target encoder and report/finding losses only; target adapters and readouts remain fixed throughout Stage 2.

All revised conditions use the same native decoder and supervised finding-head initialization, source cohort, report/finding losses, and update budgets. The finding head reads a source-prompt hidden state that cannot attend to the teacher-forced target report. Encoder and decoder LoRA parameters are independent; immutable base parameters share storage. A newly trained Stage 1 uses classification and current-report state supervision plus native current-report decoder training, with the report withheld from its image-based inputs. Its planned budget is 1,694 updates at effective batch eight; Stage 2 uses 2,400 updates at effective batch 32 on the 16,000/230/297-pair split. Physical batch and accumulation are to be fixed after 0.8B execution checks followed by 9B memory checks. The legacy final-layer-slot warm start is incompatible with this state design and is not reused. This Stage 1 subset does not establish full six-task adaptation. Revised training and evaluation remain pending; no legacy scores are reassigned to the new conditions.

The native-baseline comparison measures the entire added state branch, including its extra JEPA encoder. Planned current-slot and full-token controls with the same encoders and native decoder will distinguish the contribution of dynamics and state compression. The shuffled condition alone does not separate these effects.

BioViL-T and CheXWorld freeze their official visual encoders and train shared evidence fusion, forecast modules, and readouts with the same forecast budget. They inherit the common V-JEPA-derived Stage 1 warm start, with visual adapters reinitialized for their feature dimensions; they are adaptations, not native forecast systems from the original papers. The public VLM rows use zero-shot forecast prompts, with Qwen3.5-9B as the Qwen checkpoint. Permitted evidence and references are shared, but training regimes, pretraining data, parameter counts, and actual GPU hours differ. Nominal model sizes do not denote total system parameters. Results are single-seed point estimates; paired uncertainty estimates remain to be computed.

Copy Current repeats the source report and represents persistence for progression scoring; historical comparisons in that report are not reinterpreted as new changes after the source time. It has no continuous-score interface. The finding-transition prior fits future-finding probabilities on the 16,000 training pairs with fixed Laplace smoothing ($\alpha = 1$), conditional on the prepared source finding label and horizon, with a horizon-specific marginal fallback for unknown source labels. Its structured source annotations are a privileged source-status interface and are not VLM predictions.

093	AP, AUROC, Brier, and ECE are complete; threshold se-	143
094	lection and transition scoring remain pending. It has no	144
095	severity-direction or free-form report output.	145
096	A.2. Future-prediction metrics	146
097	Table 1 fixes four dimensions with two metrics each. The	147
098	current finding schema covers atelectasis, cardiomegaly, con-	148
099	solidation, edema, pleural effusion, and pneumothorax. A	149
100	frozen CheXbert extractor maps reference and generated	150
101	reports to findings. Reference labels that are uncertain or un-	151
102	mentioned are masked; predictions cannot reduce the evalua-	152
103	tion mask. Each metric reports its reference-defined support	153
104	and coverage, shared across methods.	
105	Future clinical status. Finding AP is the macro average	
106	of non-interpolated average precision; AUROC is also com-	
107	puted per finding and macro-averaged. Both use continuous	
108	scores and the same set of findings with positive and nega-	
109	tive reference support. Class prevalence and eligible counts	
110	accompany the results.	
111	For a zero-shot VLM, each finding is queried indepen-	
112	dently with the same source evidence and horizon. If ℓ_{yes}	
113	and ℓ_{no} are the conditional log-likelihoods of fixed candidate	
114	answers, the score is $s = \exp(\ell_{\text{yes}})/(\exp(\ell_{\text{yes}}) + \exp(\ell_{\text{no}}))$.	
115	Single-token candidates use next-token logits; multi-token	
116	candidates use complete answer log-likelihoods at a fixed an-	
117	swer boundary. Both candidates must be scored, even if one	
118	is absent from an API's top- k output. These ranking scores	
119	need not be calibrated clinical probabilities. Prompts, tok-	
120	enization, masks, and parsing are fixed before test evaluation.	
121	Trained forecasters instead use a supervised finding head; the	
122	new native supervised and state-augmented conditions will	
123	share that readout and its supervision. Comparisons with	
124	archived zero-shot Yes/No scores therefore retain a different	
125	probability source and do not isolate state conditioning.	
126	Disease progression. Transition F1 treats onset ($0 \rightarrow 1$)	
127	and resolution ($1 \rightarrow 0$) as separate events for each find-	
128	ing. Both source and future references must be assessable.	
129	Stable cases remain in the evaluation and can contribute	
130	false-positive changes; an omitted finding does not count as	
131	a correctly predicted resolution. F1 is computed for each	
132	finding-event pair and averaged over pairs with reference-	
133	positive event support. This is an event-level macro average,	
134	not a pooled micro F1 or an unweighted average of the onset	
135	and resolution subgraph means.	
136	Direction F1 assesses improved, stable, and worsened	
137	findings on reference-defined pairs where the abnormality	
138	persists at both endpoints and its direction is assessable.	
139	Annotations identify the finding, laterality, and compari-	
140	son anchored to the selected source study; conflicting or	
141	unanchored reference comparisons are ineligible. Left and	
142	right changes are scored separately when annotated, and	
	overlapping whole-patient and side-specific annotations are	
	not counted twice. F1 is macro-averaged over supported	
	finding-scope-direction classes. Missing predictions, incor-	
	rect resolutions, and parsing failures remain errors on the	
	fixed eligible references. Equal coarse severity bins alone do	
	not establish stability. The direction annotation vocabulary,	
	reference review, and supported scopes will be frozen before	
	model comparison; this label validation remains pending for	
	the 297-pair cohort. Scores from a separate 82-pair direction	
	study with a different input protocol are not inserted into the	
	main table.	
	Future report fidelity. RadGraph uses the RadGraph-	
	XL partial F1 variant, averaged over reports. GREEN [2]	
	assesses clinical content errors using a frozen evaluator.	
	The target Findings/Impression scope, scorer checkpoints,	
	prompts, and failure handling are fixed across methods be-	
	fore evaluation. These scores measure agreement with ob-	
	served follow-up reports; source-anchored changes are eval-	
	uated separately by the progression metrics.	
	Probabilistic reliability. The same future-finding scores	
	used for AP/AUROC are evaluated as probability predic-	
	tions. For finding k with fixed valid reference set $\mathcal{I}_k$, the	
	Brier score is $ \mathcal{I}_k ^{-1} \sum_{i \in \mathcal{I}_k} (p_{ik} - y_{ik})^2$, macro-averaged	
	over the common finding set. It measures overall proba-	
	bility error, including both discrimination and calibration.	
	Classwise ECE uses ten fixed equal-width probability bins	
	for each finding: it sums the absolute difference between	
	the mean predicted positive probability and observed posi-	
	tive fraction within each bin, weighted by the bin's share	
	of valid references, then macro-averages across findings.	
	Empty bins contribute zero. This adapts binned calibration	
	evaluation [1] to per-finding event probabilities rather than	
	top-label confidence. Lower Brier and ECE are better.	
	The main table evaluates uncalibrated probability predic-	
	tions. Metric definitions and archived values retain their	
	original scale; Tables 1 and 2 display scores multiplied by	
	100, except PSNR (dB), with two decimal places. Any post-	
	hoc calibration will be fitted using validation data only, with	
	a matched calibration procedure across methods, and re-	
	ported separately. Reliability diagrams, bin counts, and	
	horizon-specific comparisons remain planned supplementary	
	analyses. These metrics assess the selected follow-up cohort	
	and report-derived references, not unrestricted population	
	disease risk.	
	Supplementary analyses. The supplementary analysis	
	plan includes separate onset and resolution F1, stable-	
	reference false-change rates, direction confusion matrices,	
	horizon breakdowns, and paired patient-bootstrap intervals;	
	not all are complete. CheXbert macro-positive F1 and fu-	
	ture retrieval remain supplementary. Planned Future R@1	

ranks the actual follow-up and fixed other-patient candidates by disagreement with a common predicted finding schema, extracted from reports or emitted by the transition prior. Candidates must be matched on horizon, acquisition conditions, coarse current findings, and reference assessability; ties split first-place credit equally. Candidate-pool size and eligible-query counts will accompany retrieval scores. Degree errors require reviewed ordinal annotations. These analyses do not add columns to the fixed eight-metric table.

A.3. Downstream supervision

The revised Stage 1 design uses eight 1024-dimensional slots: four from vision–language fusion layers and four from the VLM’s native vision encoder. Each source contributes an early layer, two intermediate layers, and its final layer through learned aggregation and projection. Classification and disease recognition read all eight slots; segmentation and SR read only $S_{5:8}$ together with the task image, which is LR for SR. Task gradients jointly train the slot readouts and decoders. This construction differs from the initial last-layer-slot prototype and requires new training. The expanded six-task evaluation adds standard VQA, report generation, and phrase grounding with the same-exam report withheld.

Classification uses 13 report-derived finding/device labels, excluding No Finding, and masks uncertain or missing labels. The test set has 353 images; macro ranking metrics include the 11 labels with positive and negative support. DINOv2 and CheXWorld classification heads use 13,681 training images, 160 validation images, and 20 epochs. Their stored `macro_auprc` field implements non-interpolated average precision. Current-report scoring uses 507 images and withholds the target Findings/Impression. Standard VQA is planned on MIMIC-CXR-VQA with its full answer ontology, negative examples, and empty answer sets. Official Accuracy and micro F1 will be supplemented by question-type and region breakdowns. MS-CXR phrase grounding requires fixed heatmap-to-box conversion, multi-box matching, and invalid-output treatment. Derived disease-list QA and Chest ImaGenome anatomical-region localization are not substitutes for those benchmarks. Benchmark test patients must be excluded from all task and stage training sources; available public-checkpoint pretraining overlap remains to be documented.

Segmentation uses quality-filtered CXAS pseudo-labels for the two lungs and heart, with 4,096/249/447 training/validation/test images. Human Dice separately measures the two lungs on 138 Montgomery images, without a heart label. SR uses the same CXR split and antialiased bicubic downsampling by four per spatial dimension, with HR long edge capped at 512 pixels. Every image input receives LR for SR; HR supplies supervision only. Padding is excluded from losses and image metrics. The dense-task rows use 20 epochs, effective batch size eight, and seed 20260913.

Qwen3.5-9B visual-slot rows freeze the native vision encoder and train a task decoder conditioned on four multi-depth visual slots. Their shuffled control permutes these conditions across patients; the image-only decoder uses zero context. These rows do not use the eight-slot forecast pilot or full language model. DINOv2/CheXWorld freeze spatial encoders and train the same pixel-task decoder; their spatial-token counts differ from the four-slot interface. SwinIR-M uses public DF2K $4\times$ pretraining and adapts all 11.9 million parameters to the CXR degradation. Matching data and epochs does not match compute: SwinIR consumes 8.313 GPU hours including evaluation, versus 0.161 for the image-only SR run on the same GPU model. MAIRA-2 retains its native report/grounding interface, but its results are pending.

The planned no-slots baseline and full MedWorld-JEPA will share six-task supervision and decoder budgets; those rows remain TBD. Earlier report-assisted scores, $2\times$ SR, and separate two-epoch joint dense-task pilots are excluded. [TBD: complete full six-task adaptation, official VQA and MS-CXR grounding, and cross-task exclusion/pretraining-overlap audits for Table 2].

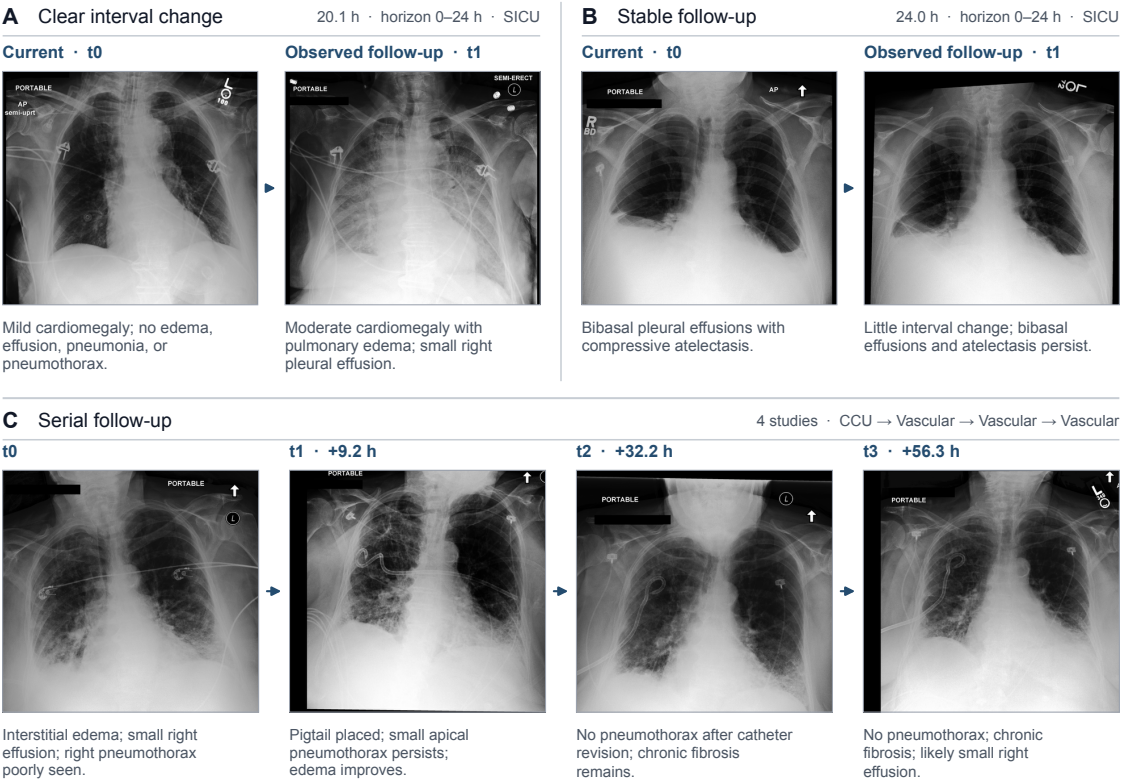


Figure S1. Representative longitudinal MIMIC-CXR trajectories. Each row shows the current radiograph, elapsed time and horizon bin, and the observed follow-up. Cases A and C exhibit clear interval change, whereas Case B is a stable follow-up. These examples illustrate the longitudinal pairs and are not model predictions.

266 **B. Candidate Visual Analysis**



Figure S2. Planned patch weights from actual slot-readout attention, grouped by source and depth. This panel requires an attention-based readout; slots have no prescribed anatomical assignments.

267 **References**

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